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10 December 1984
SUPERSEDING
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MILITARY SPECIFICATION

CUTOFF MACHINES, ABRASIVE DISK OR CIRCULAR SAW BLADE, METALCUTTING, CHOP STROKE, DRY CUT, FLOOR MOUNTED

This specification is approved for use by all Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers metalcutting, chop stroke, cutoff machines of the types, classes and sizes classified in 1.2.

1.2 Classification. This specification covers machines of the following types, classes and sizes. The type, class and size required for specific procurements shall be as specified (see 6.2.1).

Type I - Manual and Power Operated Head

Class I - Fixed, Non-Swiveling Head
- Size 12 - 12 inch wheel - (304.80mm)
- Size 14 - 14 inch wheel - (355.60mm)
- Size 16 - 16 inch wheel - (406.40mm)
- Size 20 - 20 inch wheel - (508.00mm)
- Size 24 - 24 inch wheel - (609.60mm)
- Size 26 - 26 inch wheel - (660.40mm)
- Size 30 - 30 inch wheel - (762.00mm)

Class II - Swiveling Head
- Size 12 - 12 inch wheel - (304.80mm)
- Size 14 - 14 inch wheel - (355.60mm)
- Size 16 - 16 inch wheel - (406.40mm)
- Size 20 - 20 inch wheel - (508.00mm)
- Size 24 - 24 inch wheel - (609.60mm)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3405

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Type II - Automatic

Class I - Fixed, Non Swiveling Head
Size 14 - 14 inch wheel - (355.60mm)
Size 16 - 16 inch wheel - (406.40mm)
Size 20 - 20 inch wheel - (508.00mm)
Size 24 - 24 inch wheel - (609.60mm)

Class II - Swiveling Head
Size 14 - 14 inch wheel - (355.60mm)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARD

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI B.11.10 - Safety Requirements for the Construction, Care, and Use of Metal Sawing Machines.

ANSI/NFPA 79 - Metalworking Machine Tools and Plastics Machinery, Electrical Standards for.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-3951 - Commercial Packaging, Standard Practice for.

ASTM-E-380 - Metric Practice Guide.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the test of this specification shall take precedence.

3. REQUIREMENTS

3.1 First Article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Design. The machine shall be new and one of the manufacturer's current models capable of cutting ferrous or nonferrous bar stock in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in ASTM-E-380.

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3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the US Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, pressure gauges, indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1.1 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI system of measurements.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The machine and all its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with operation of the machine. The safety devices shall prevent unintentional contact with the guarded part and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4). In addition, the machine shall comply with all requirements of ANSI B11.10 that are designated therein as the responsibility of the machine manufacturer. In the event of a conflict between the requirements of the OSHA and ANSI Standards, the requirements of OSHA shall apply.

3.2.6 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings and filler caps shall be accessible.

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3.2.7 Interchangeability. To provide for replacement of worn or damaged parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of the replacement parts without modification to the part or the machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scales and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. All surfaces to be painted shall be cleaned of all foreign matter. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with manufacturer's standard commercial practice which shall be not less than one coat of primer and one coat of finish color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose. All gears shall be machined and have a surface finish and toughness that will enable the gear train to transmit full-rated torque of the drive motor without gear damage, failure or premature deterioration and wear. All gears shall be manufactured to definite established standards to provide for interchangeability and parts replacement.

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3.4 Components.

3.4.1 Type I machines.

3.4.1.1 Components for type I machines. Type I machines shall consist essentially of a base, workholder, and a wheelhead having a spindle that will accommodate either an abrasive disk (for cutting ferrous metals) or a steel circular saw blade (for cutting nonferrous materials). The components for type II machines shall conform to the requirements of 3.4.1.1.1 through 3.4.1.1.5.

3.4.1.1.1 Base. The base of type I machines shall be either a casting or a steel weldment capable of rigidly supporting the workholder and wheelhead at a convenient operating height. The base shall have means for securely anchoring the machine to the floor.

3.4.1.1.2 Workholder. Type I machines shall have a workholder that will rigidly hold all regular shapes of bar stock of the sizes specified in table I. Unless otherwise specified (see 6.2.1), the workholder may be of either the vise, fence, V-block or L-block type and shall have means for secure clamping of the workpiece. Unless otherwise specified, the clamping device may be either hand operated or foot operated. When specified (see 6.2.1), the clamping device shall be pneumatically operated and shall conform to the requirements of 3.4.1.1.2.1. The workholder for all machines shall be capable of securely holding the workpiece in the proper position for 90-degree cuts. When specified (see 6.2.1), the workholder shall swivel and shall be capable of holding the workpiece in the proper position for any angular cut from 45 degrees to 90 degrees of the workpiece axis. Workholders shall have a suitable clamp that will securely hold the workholder at any angle within its adjustment range.

3.4.1.1.2.1 Pneumatically operated work clamping device. When a pneumatically operated work clamping device is required (see 3.4.1.1.2 and 6.2.1), the device shall be capable of securely clamping regular shaped bar stock up to the machine's rated capacity when operating on a 90 to 110 PSI air supply. The pneumatic system shall include an air filter, pressure regulator, operating valve, and all necessary piping, hose, tubing, and fittings.

3.4.1.1.3 Wheelhead assembly. The wheelhead shall be of the chopstroke type and shall consist essentially of a wheelhead arm, spindle, spindle drive motor, spindle drive system, counterbalance, and guards for the disk or blade. The wheelhead assembly of class I machines shall be of the fixed, nonswiveling type intended primarily for 90-degree cutoff operations (angular cuts shall be possible, however, using one of the optional swiveling type workholders described in 3.4.1.1.2). The wheelhead of class II machines shall be of the type in which the entire workhead can be swiveled to cut any angle from 45 degrees to 90 degrees of the workpiece axis without

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the use of a swivel type workholder. The swiveling head shall have a suitable clamp that will securely hold the head at any angle within its range of adjustment.

3.4.1.1.3.1 Arm. The arm shall pivot on a trunnion which is securely mounted on the upper surface of the base. The trunnion bearings shall be sealed to prevent loss of lubricant and the entrance of metal cuttings, abrasive dusts, or other extraneous materials. The arm shall be adequately rigid to support the spindle, drive motor, drive mechanism, guards and other parts mounted thereon. Unless otherwise specified (see 6.2.1), the arm shall be manually operated and shall have an operating handle or lever located convenient to the operator. When specified (see 6.2.1), the arm shall be operated by either a hydraulic or pneumatic power system. If a hydraulic system is furnished, it shall be complete and shall consist of a motor driven hydraulic pump, relief valves, piston and cylinder assembly and all necessary linkage, controls, piping, tubing, hoses, and fittings. If a pneumatic system is furnished, it shall be capable of operating on a 90 to 110 PSI air supply and shall include an air filter, pressure regulator, operating valve, and all necessary piping, hose, tubing, and fittings.

3.4.1.1.3.2 Spindle. The spindle shall be mounted on anti-friction bearings of a size and type suitable for the application. The bearings shall be sealed to prevent loss of lubricant and the entrance of metal or abrasive particles or other extraneous matter. The spindle shall be driven by an electric motor conforming to 3.4.2.1.9 and a drive system providing the disk speed specified in table I. The spindle shall be designed to accommodate abrasive disks and circular saw blades conforming to 3.4.1.1.3.3 and shall have means for secure and safe mounting of the disk and blade. A pair of abrasive disk flanges shall be furnished.

3.4.1.1.3.3 Abrasive disk and circular saw blade. The machine shall be designed to utilize an abrasive disk for cutting ferrous metals and a circular steel saw blade for cutting nonferrous materials. The disks and blades shall be capable of sawing at a rate that will meet the test requirements specified in 4.6. When required (see 6.2.1), the specified quantity and type of abrasive disks or circular steel saw blades shall be furnished.

3.4.1.1.3.4 Wheel guards. The machine shall have a wheel guard that will protect the operator from sparks and abraded material. The guard shall have sufficient strength to provide operator protection in the event of breakage of the abrasive disk. The guard shall not interfere with the operation of the machine and shall be removable to facilitate cleaning and servicing of the machine.

3.4.1.1.3.5 Counterbalance. Machines with a manually operated wheelhead arm shall have a counterbalance system that will return the wheelhead to the raised position when the operating lever is released.

3.4.1.1.3.6 Wheelhead stop. If the design of the machine is such that overtravel of the wheelhead would cut the base or bed, the machine shall

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have an adjustable stop that will limit the travel of the wheelhead to prevent such damage.

3.4.1.1.4 Length gauge. Unless otherwise specified (see 6.2.1), type I machines shall have a length gauge to facilitate cutting multiple pieces to a predetermined length. For size 12 machines, the length gauge shall be adjustable to cut lengths ranging from one inch, or less, to 18 inches, or more. The length gauge shall not interfere with the cutting action of the machine.

3.4.1.1.5 Mist spray system. When specified (see 6.2.1), type I machines shall have a mist spray system suitable for use in cutting nonferrous metals using circular steel saw blades.

3.4.1.1.6 Oscillating (reciprocating) mecharism. When specified (see 6.2.1), type I machines shall have a mechanism that will oscillate (reciprocate) the head and abrasive disk in a direction parallel to the kerf in the workpiece.

3.4.2 Type II machines.

3.4.2.1 Components for type II machines. Type II machines shall have components similar to those on type I machines and shall in addition have all components necessary for automatically cutting multiple pieces to a predetermined length. The components for type II machines shall conform to the requirements of 3.4.2.1.1 through 3.4.2.1.7.

3.4.2.1.1 Automatic mechanism. Automatic machines shall have an automatic bar feed mechanism, a pneumatically or hydraulically operated workholder and workhead, and all necessary automatic controls required to produce the following automatic action. The machine shall be automatic to the extent that after the rear bar stock is manually loaded into the bar feeder and after the length adjustment has been performed, the bar feeder shall automatically feed downward to cut the workpiece. After cutting the stock, the workhead shall retract to the raised position and the cycle shall be automatically repeated until the bar feeder reaches the end of its travel at which time the machine shall automatically stop in a position ready for manually loading another length of bar stock into the feeder.

3.4.2.1.2 Base. The base of type II machines shall conform to the requirements of 3.4.1.1.1. In addition, the base shall have a bar feeder track and a discharge chute for discharging the cut pieces into a container (container required or normally furnished). The feeder track and discharge chute may be either an integral part of the base or separate assemblies securely attached thereto.

3.4.2.1.3 Automatic bar feeder mechanism. The automatic bar feeder mechanism shall be capable of feeding regular shaped stock within the size capacity specified in table I and having an initial length not less than 12 feet. Unless otherwise specified (see 6.2.1), the stroke of the feeder

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shall be adjustable to cut lengths over a range of 1/2-inch, or less, to eight inches, or more.

3.4.2.1.4 Workholder. The workholder of type II, class I machines shall be of the fixed, nonswiveling, pneumatically operated type conforming to the applicable requirements of 3.4.1.1.2 and 3.4.1.1.2.1.

3.4.2.1.5 Wheelhead. The wheelhead of type II, class I machines shall be of the fixed, non-swiveling, pneumatically or hydraulically actuated type conforming to the requirements of 3.4.1.1.3. The wheelhead of type II, class II machines shall be of the swiveling, pneumatically or hydraulically actuated type conforming to the requirements of 3.4.1.1.3. The arm, hydraulic or pneumatic system, spindle, abrasive disk or circular steel saw blade, and guard shall conform to the applicable requirements of 3.4.1.1.3.1 through 3.4.1.1.3.6.

3.4.2.1.6 Mist spray system. Type II machines shall have a mist spray system conforming to 3.4.1.1.5.

3.4.2.1.7 Oscillating (reciprocating) mechanism. When specified (see 6.2.1), type II machines shall have an oscillating (reciprocating) mechanism conforming to 3.4.1.1.6.

3.4.2.1.8 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA 79. Each machine shall draw all of its electrical power from a single 230/460 volt, 3 phase, 60 Hz circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 230 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA 79 for the disconnect fuse rating.

3.4.2.1.9 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a dripproof enclosure.

3.4.2.1.10 Solid state components. Solid state design shall be used throughout for electronic components. The use of selenium or other similar ageing devices is permitted only for voltage surge protection to other solid state components.

3.4.2.1.11 Control circuit voltage. Auxiliary control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 115 volts.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and characteristics of the machine shall be not less than those specified in table I for the machine size specified.

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CHARACTERISTIC	TYPE I (Manual and Power Head)												TYPE II (Automatic)				
	CLASS I (Fixed Head)							CLASS II (Swiveling Head)					CLASS I (Fixed Head)				CLASS II (Swivel Head)
	12	14	16	20	24	26	30	12	14	16	20	24	14	16	20	24	14
Abrasive Disk and Circular Saw Blade:																	
Diameter, inches	12	14	16	20	24	26	30	12	14	16	20	24	14	16	20	24	14
Width (Abrasive disk), inch	3/32	1/8	1/8	1/8	5/32	7/32	7/32	3/32	1/8	1/8	1/8	5/32	1/8	1/8	1/8	5/32	1/8
Hole Size, inches	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Speed, Surface feet per minute	← 10,000 - 14,000 →							← 10,000 - 14000 →					← 10,000 - 14,000 →				10,000 - 14,000
Cutting Capacity, at 90° Cut-off,																	
Ferrous:																	
Round Solids, O.D., inches	1 1/4	1 1/2	2	2 1/2	4	6	8	1 1/4	1 1/4	2	2 1/2	4	1	1 1/4	1 3/4	2	1 1/2
Round Tubing, O.D., inches	2	2 1/2	3	5 1/2	6	8	8	2 1/4	2 1/4	3	5	6	2	2 1/2	3 1/2	4 1/2	2 1/2
Drive Motor Horsepower																	
Standard	3	5	7 1/2	10	15	20	25	3 1/2	3	5	10	15	3 1/2	7 1/2	10	15	3 1/2
Optional	-	3 1/2 or 7	5 or 10	15 or 20	25	25	-	-	5	7 1/2	15 or 20	-	7	-	15	-	-

Requirements shall be not less than those shown for the type, class and size machine ordered and not greater than the minimum requirements of the next size larger. If the largest size offered is ordered, the maximum requirements shall not exceed 10 percent of stated requirements.

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3.6 Performance and accuracy. The machine shall be capable of meeting the performance and accuracy requirements of 4.6.2.

3.7 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine:

a. One set of hand tools normally furnished by the manufacturer.

b. One abrasive disk (the disk used in performance testing of the machine).

3.8 Optional equipment. Optional equipment and accessories shall be furnished as specified (see 6.2.1).

3.9 Marking of instruction plates, indicating plates and nameplates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

3.9.1 Lubrication chart plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.9.2 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.10 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract.

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3.11 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of the item to pass the first article tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of at least the examination in 4.5 and the tests in 4.6.1. Failure of the item to pass the examination or tests shall be cause for rejection.

4.5 Examination. The machine and equipment shall be examined to determine compliance with the requirements of 3.2 through 3.11.

4.6 Tests.

4.6.1 Operational test. The machine shall be operated at no load for not less than 15 minutes for type I machines and not less than 30 minutes for type II machines. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period. The automatic operating sequence of type II machines shall conform to the automatic operation requirements of 3.4.2.1.1.

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4.6.2 Performance and and accuracy test.

4.6.2.1 Preparation for tests. All tests shall be performed using an SAE 1020 solid round steel bar having a nominal diameter equal to the maximum rated capacity of the machine and using an abrasive disk of a size conforming to table I.

4.6.2.2 Cutting rate test. Type I machines shall cut three disks, not less than one inch thick, from the workpiece. Type II machines shall cut five disks, not less than one inch thick, from the workpiece using the automatic mechanism. All cuts shall be at a 90 degree angle. The actual cutting time shall be observed and the cutting rate shall be calculated. The cutting rate for all machines shall be not less than 15 square inches per minute. There shall be no evidence of damage (wear excepted) to the abrasive disk.

4.6.2.3 Accuracy test. The last cut in the test specified in 4.6.2.2 shall produce a piece not less than four inches long. The cuts on the four-inch piece shall be square with the axis of the piece within $\pm$0.005 inch per inch of diameter. The four equal length pieces produced on type II machines shall be uniform in length within $\pm$0.010 inch.

4.6.2.4 Angular cuts. All machines having a swivel workholder or a swivel head shall cut one piece with the workholder or head, as applicable, swiveled to the proper position to cut a 45 degree angle. The cut face of the sample shall be within $\pm$1/2 degree of the desired 45 degree angle.

4.7 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging, packing and marking. Unless otherwise specified, packaging, packing, and marking shall be in accordance with ASTM-D-3951. When specified (see 6.2.1), level A packaging, level A or level B packing and special marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. Machines covered by this specification are intended to be used in cutoff operations on ferrous metals, nonferrous metals, and nonmetallic materials, in the form of solid bars, tubing, pipe, and extrusions.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class and size required (see 1.2).

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c. Specify when first article is required for inspection and approval (see 3.1).

d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Dual calibrations (US and SI), if required (see 3.2.1.1.1).

g. Additional safety and health requirements, if required (see 3.2.5).

h. Painting, if different (see 3.3.5).

i. Specify type of workholder (see 3.4.1.1.2)

j. Pneumatically operated workholder, if required (see 3.4.1.1.2).

k. Swivel workholder, if required (see 3.4.1.1.2).

l. Power operated wheelhead arm, if required (see 3.4.1.1.3.1).

m. Quantity and type of abrasive disks or circular steel blades, if required (see 3.4.1.1.3.3).

n. Length gauge, if not required (see 3.4.1.1.4).

o. Mist spray system, if required (see 3.4.1.1.5).

p. Oscillating head mechanism, if required (see 3.4.1.1.6 and 3.4.2.1.7).

q. Bar feeder stroke range, if different (see 3.4.2.1.3).

r. Electrical system and input power supply, if different (see 3.4.2.1.8).

s. Motor enclosure, if different (see 3.4.2.1.9).

t. Size and capacity, if different (see 3.5).

u. Standard equipment, if different (see 3.7).

v. Specify and fully describe optional equipment, if required (see 3.8).

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w. Lubrication chart or plate, if different (see 3.9.1).

x. Nameplate, if different (see 3.9.2).

y. First article inspection, if different (see 4.3).

z. Quality conformance inspection, if different (see 4.4).

aa. Level of packaging, packing and marking, if different (see 5.1).

6.2.2 Contract data requirements. All data requirements, such as parts lists, wiring diagrams, operating manuals, maintenance manuals, alignment and test results, shall be requested on a DD Form 1423, Contract Data Requirements List and shall be furnished in accordance with each specified DD Form 1664, Data Item Description as shown on the DD Form 1423.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, heat, etc. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machines in the environment) the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. The machine shall be equipped with all points-of-operation guarding normally furnished as standard on the manufacturer's commercial machines supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required, as in most cases, the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements, should be specified in detail under 6.2.1.

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6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Preparing Activity:
DLA-IP

Project Number:
3405-0074

Review Activities:
Army - AL
Navy - SH
Air Force - 84
DLA- GS

User Activities:
Army - ME
Navy - MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-80021B	Cutoff Machines, Abrasive Disk or Circular Saw Blade, Metalcutting, Chop Stroke, Dry Cut, Floor Mounted

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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